

RITESH PATIL

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SUMMARY:

Machine Learning Engineer with experience in Generative AI and Computer Vision, building RAG-based chatbots and YOLO object detection systems. Skilled in Python, LLMs, TensorFlow, OpenCV, and vector databases (FAISS, Pinecone). Experienced in developing end-to-end ML pipelines and deploying scalable AI solutions using AWS, Docker, and CI/CD.

EDUCATION:

P.G Diploma in (AI & ML) **MITWPU** July 2024 – May 2025

- Relevant Coursework: Machine Learning, AI, Data Mining, Neural Networks, NLP, Computer Vision, Statistical Analysis, Deep Learning: applied concepts through hands-on projects

BSc Computer Science **Abasaheb Garware college** Aug 2020 – June 2023

- Computer Networks, Database Systems, Algorithms, Programming Languages, Computer Architecture, and AWS Cloud

LANGUAGES AND TECHNOLOGIES:

Languages : Python , C , R, SQL

Libraries : Pandas, Numpy, Scikit-learn, openCV, Matplotlib, Seaborn, LangChain, LangGrap, FastAPI

AI/ML skill: Generative AI, Agentic AI, LLM, RAG, image processing, Ollama (Local LLM)

AWS : AWS (EC2, S3, Lambda, IAM, VPC), Docker, CI/CD

PROFESSIONAL EXPERIENCE:

Machine Learning engineer **Automation Teknix** Nov 2025- Mar 2026

- Built an end-to-end industrial object detection system using YOLO, covering data annotation, preprocessing, augmentation, model training, and evaluation on AWS SageMaker for defect and component detection
- Converted the trained YOLO model to TensorFlow Lite and integrated it into an offline Android app, ensuring smooth and accurate real-time object detection on-device

Software engineer (Intern) **Jnana Prabodhini** Jan 2025 - June 2025

- Contributed to developing both offline and online versions of the [Virtual Kiosk](#) using AngularJS, enabling content delivery through smart JSON routing with .exe deployment and YouTube fallback
- Engineered a responsive UI using core AngularJS principles enhancing kiosk interactivity and user engagement across platforms

PROJECTS:

Multimodal RAG Chatbot – Ollama

- Built a Multimodal RAG system using Ollama (llama2:7b), nomic-embed-text embeddings, and FAISS for accurate semantic retrieval from PDFs and OCR data.
- Implemented persistent conversational memory (LangGraph + SQLite) to maintain context and reduce hallucinations through strict grounding
- Designed a scalable, modular RAG architecture optimized for efficient local LLM inference and real-world deployment

Medical Chatbot (RAG, LangChain, GPT-4, Pinecone, AWS, Docker)

- Built end-to-end RAG-based chatbot using 637-page medical textbook with Sentence Transformer embeddings
- Orchestrated LangChain, GPT-4, and Pinecone for retrieval-augmented, context-aware medical query resolution
- Implemented CI/CD pipeline with GitHub Actions, containerized with Docker, and deployed on AWS EC2

Heart Disease Prediction Using Retina Images (YOLO, TensorFlow, OpenCV)

- Implemented YOLO for precise detection and classification of retina features linked to heart disease, improving detection
- Utilized Roboflow for annotating 800+ images and training data, reducing preprocessing time by 50%
- Designed and trained CNNs in TensorFlow and OpenCV, increasing marker identification accuracy by 25%

CERTIFICATION:

[Oracle Cloud Infrastructure 2025 Certified Generative AI Professional](#) | LLM, RAG

[GenAI & LLM Workshop](#) | LMStudio, AnythingLLM, RAG, HuggingFace, Prompting Strategies